Sharpening Where There Is Headroom: What GRPO Changes in Text-to-SQL Across Model Scale

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Code and results: https://github.com/ved-2004/RLVR-Text-to-SQL

1 Introduction and Motivation

Reinforcement learning from verifiable rewards (RLVR) has become a standard post-training step for reasoning and code models [2, 5, 9]. The recipe is simple: sample several answers, check each with a program, and push the model toward the ones that pass. What is less clear is *what* this changes. A model that gets more questions right after RL could have learned to solve new problems, or it could have learned to pick, on the first try, an answer it was already able to produce some of the time. Yue et al. [12] argue that on math and code benchmarks it is mostly the second: RL raises pass@1 but does not raise, and can lower, pass@k for large k.

This report studies that question on text-to-SQL, which is a useful instrument for three reasons. First, the verifier is cheap and objective: execute the query and compare result sets. Second, the verifier has a known and measurable false-positive rate, because a wrong query can return the right rows on one database by coincidence [13]. This lets us check whether the policy learns to exploit the reward. Third, SQL tokens carry explicit roles (keywords, aggregates, table names, aliases, literals), so token-level changes can be grouped by what they do.

We train Qwen2.5-Coder-Instruct at 1.5B and 7B parameters [4] with one fixed GRPO recipe and a binary execution reward on the Spider training set [11], then ask four questions:

1. How much does the same recipe help at each scale, and what predicts the size of the gain?
2. Where in the output distribution does the model move, and does that differ by scale?
3. Does RL change *how* the model writes SQL, not only whether it is correct?
4. Does the policy learn to exploit the verifier's false positives?

Our main findings are:

- The same recipe raises execution accuracy by +12.7 pp at 1.5B (59.1% to 71.8%) and by +2.6 pp at 7B (76.8% to 79.4%). The size of the gain lines up with the base model's pass@1 to pass@8 gap (18.8 vs. 6.8 points). This is a two-point observation, not a fitted law.
- The 1.5B model moves 3.65× further from its base in per-token KL and fixes 155 dev questions versus 35 for the 7B. The 1.5B change is broad; the 7B change is concentrated on aggregate and `FROM`-clause tokens.

- Without any style term in the reward, the 1.5B model drops table aliases (2.63 to 0.36 per query) and switches to `table.column` references, which is how the untrained 7B already writes. It moves *away* from the style of the gold queries.
- The single-database vs. test-suite gap stays between 6 and 9 points. It narrows at 7B and widens slightly at 1.5B, so we see no clear sign of reward hacking, but also no sign that RL removes the false positives.

2 Background and Related Work

GRPO and RLVR. Group Relative Policy Optimization [9] drops the value network of PPO. For each prompt it samples a group of G completions, scores them, and uses the group-normalized reward as the advantage:

$$\hat{A}_i = \frac{r_i - \text{mean}(r_1, \ldots, r_G)}{\text{std}(r_1, \ldots, r_G)}. \tag{1}$$

A KL penalty with coefficient β keeps the policy near a reference model. When every completion in a group gets the same reward, all advantages are zero and the prompt contributes no gradient. DAPO [10] addresses this with dynamic sampling, which filters out such groups. RLVR with GRPO-style updates is used in DeepSeek-R1 [2] and Tülu 3 [5].

Does RL add capability? Yue et al. [12] compare base and RL-trained models with pass@k [1] at large k and find that base models match or beat RL models once k is large, which suggests RL mostly reweights solutions already in the base model's distribution. Mukherjee et al. [7] find that RL updates only a small subnetwork of parameters, consistent with a light-touch change. Our scale comparison asks a related question from a different side: if RL mostly converts existing pass@k into pass@1, the room available to convert should predict the gain.

RL for text-to-SQL. SQL-R1 [6] trains reasoning models with GRPO and execution-based rewards. Reasoning-SQL [8] adds partial rewards for schema linking, syntax, and n-gram similarity. Both target state-of-the-art accuracy. We instead keep the recipe minimal (binary reward, no reasoning trace, LoRA) so that changes can be attributed and compared across scale.

Evaluating SQL by execution. Spider [11] reports exact set match and execution accuracy. Execution accuracy has false positives: a query such as `WHERE age > 34` may return the same rows as the gold `WHERE age >= 34` on one database. Test-suite accuracy [13] runs each query on a distilled set of perturbed databases and counts it correct only if it matches gold on all of them. Because our reward is execution on a single database, the gap between single-database and test-suite accuracy measures how much of the reward is earned by coincidence.

3 Experiments and Results

3.1 Setup

Models. Qwen2.5-Coder-1.5B-Instruct and Qwen2.5-Coder-7B-Instruct [4]. These are general code models, not SQL-specialized, so that a gain from RL is not confounded with prior SQL fine-tuning.

Training. GRPO via TRL `GRPOTrainer` with LoRA [3] ($r = 16$, $\alpha = 32$, all linear layers), $G = 8$ completions per prompt, learning rate 10^{-5}, $\beta = 0.04$, sampling temperature 0.9. Training data is the full Spider train set (7,000 examples). [TODO: epochs, batch size, max completion length, wall-clock and GPU type]. The prompt gives the schema as one line per table and asks for a single SQLite query in a `sql` code block. There is no reasoning trace.

Reward. Binary execution reward: extract the query, run it read-only on the example's SQLite database with a 5 s timeout, and return 1 if its result set equals the gold result set, else 0. Write statements are rejected.

Evaluation. Full Spider dev set (1,034 questions), greedy decoding with vLLM, identical prompt for all four models. We report:

- **EX**: execution accuracy from the test-suite evaluator [13] run on the original dev databases.
- **TS**: the same evaluator run on the distilled test suite. Because only the databases change, **EX** − **TS** (the FP gap) isolates false positives.
- **EX (official)**: the original Spider evaluator, for comparison.
- **pass@1 / pass@8**: from 8 samples per question, scored with a set-based result comparison. [TODO: sampling temperature for pass@k].
- **Set-op EX**: execution accuracy on the ?? dev questions whose gold query uses `INTERSECT`, `UNION` or `EXCEPT`.
- **ECE**: expected calibration error. [TODO: state exactly what confidence is binned].

Evaluation note. Two earlier versions of these numbers were wrong. The first pipeline truncated long prompts at 1,536 tokens and reported a 1.5B base EX of 34.5%. The second saved only the last 2,500 characters of the official evaluator's log, which cut off the execution row; the value we read as EX was the SELECT-clause partial-match recall. All numbers in this report are generated from the stored result files by `report/make_numbers.py`, which reads each metric by its row in the evaluator output.

3.2 Main results

GRPO helps the 1.5B model on every execution metric by about 12 to 13 points and the 7B by about 2 to 3.5 (Table 1). The untrained 7B still beats the trained 1.5B on EX (0.768 vs. 0.718). Calibration gets slightly worse at both scales.

3.3 The gain tracks sampling headroom

In both models pass@1 rises more than pass@8 (Table 2, Figure 1), and the EX gain closely matches the pass@1 gain. Most of the improvement is turning questions the model sometimes solved into questions it solves on the first try. The model with more of these questions available (1.5B) gains more.

This is not shown to be pure sharpening: pass@8 also rises at both scales. Whether the 1.5B learns set operations it could not do before is measured by the set-op row of Table 1. With two models and one seed, we read Table 2 as consistent with "gain scales with headroom," not as evidence for it.

Table 1: Spider dev results. Δ is in percentage points.

	1.5B			7B		
Metric	Base	GRPO	Δ	Base	GRPO	Δ
EX	0.591	0.718	+12.7	0.768	0.794	+2.6
TS	0.510	0.632	+12.2	0.695	0.730	+3.5
FP gap (pts)	8.1	8.6	+0.5	7.3	6.4	−0.9
EX (official)	??	??	??	??	??	??
pass@1	0.566	0.689	+12.3	0.729	0.755	+2.6
pass@8	0.753	0.786	+3.3	0.797	0.816	+1.9
Set-op EX	??	??	??	??	??	??
ECE	0.049	0.062		0.074	0.086	

Table 2: Headroom before training vs. gain after training (points).

Model	pass@8 − pass@1 (base)	Δ pass@1	Δ pass@8	Δ EX
1.5B	18.8	+12.3	+3.3	+12.7
7B	6.8	+2.6	+1.9	+2.6

3.4 Where the model moves: token-level divergence

We teacher-force the gold query through each base and GRPO model and compute, per token, the KL divergence between their next-token distributions, the change in entropy, and the change in gold-token log-probability. Tokens are tagged by SQL role (keyword, aggregate, table, column, alias, literal, punctuation).

- **Magnitude.** Mean per-token KL is 0.043 nats at 1.5B and 0.012 at 7B, a 3.65× difference. GRPO fixes 155 dev questions at 1.5B (wrong before, right after) and 35 at 7B. [TODO: report questions broken (right before, wrong after) at each scale].
- **Shape at 1.5B.** Entropy drops by about 0.11 nats with no strong preference for any token role.
- **Shape at 7B.** On the questions GRPO fixed, KL is concentrated on aggregate tokens (17.3× their share of tokens) and on `FROM`-clause tokens (5.2×).

One reading is that the 1.5B model is still learning general habits for this task, while the 7B already has them and only adjusts a few decisions.

3.5 Surface-form convergence

The reward only checks the result set. Even so, GRPO drives the 1.5B model to drop the alias convention used in Spider's gold queries and to write qualified `table.column` references instead, close to how the untrained 7B already writes. The 7B's style does not change. The 1.5B also uses fewer joins, which matters because unneeded joins are a common error for this model. Whether this style shift *causes* part of the gain is tested in E3 and E4 (Section 3.9).

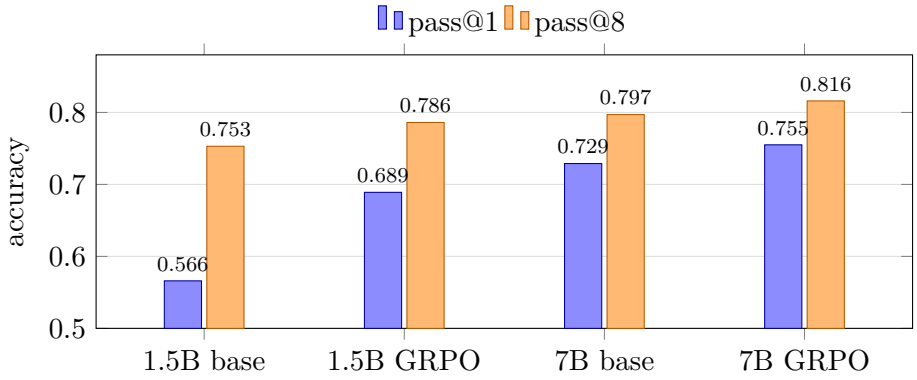


Figure 1: pass@1 rises much more than pass@8, and the rise is largest where the gap between them was largest.

Table 3: Surface style of generated SQL, per query on Spider dev.

	Gold	1.5B base	1.5B GRPO	7B base	7B GRPO
Aliases (`T1`, `T2`, ...)	2.93	2.63	0.36	0.43	0.45
Qualified `table.column`	[TODO:]	0.46	1.41	1.62	1.59
`JOIN`	[TODO:]	0.51	0.35	[TODO:]	[TODO:]

3.6 False positives and reward hacking

A binary execution reward on one database pays for coincidentally correct queries, so a policy could learn to produce them. If it did, the EX − TS gap should grow during training. It moves by +0.5 points at 1.5B and −0.9 at 7B (Table 1). Within this recipe we see no strong reward hacking, but the false-positive rate is large enough that it should be reported next to any execution-accuracy gain.

3.7 Training dynamics

A mean of 0.625 of GRPO groups had zero reward variance (`frac_reward_zero_std`): all 8 completions were right, or all were wrong, so they gave no gradient. Most of the compute went to prompts that were already solved or still out of reach. This is the case dynamic sampling in DAPO [10] is designed for.

3.8 Limitations

Two model sizes, one seed per configuration, one dataset, one reward. The headroom observation rests on two points. pass@k uses a looser result comparison than the evaluators. LoRA limits how far the policy can move, so some differences between scales could come from LoRA capacity rather than RL itself.

3.9 Running and next experiments

1. **E3, is the style shift causal? (running)** Prompt the 1.5B base and GRPO models to write with or without aliases, and add execution-voted sampling (8 samples, majority vote

on result sets, no gold). If forcing the base model to write `table.column` recovers part of the gain, style is part of the mechanism. If voting matches GRPO, RL here is mostly selection.

2. **E4, which neurons carry the style? (running)** Rank all 250,880 MLP neurons of the 1.5B by their effect on alias preference, using gradients and attribution patching from base to GRPO activations. Steer the top neurons and measure EX and alias rate on dev, with random-neuron controls and the reverse direction on the GRPO model.

3. **Robustness.** Two more seeds at 1.5B, and DAPO at 1.5B to test whether removing zero-variance groups changes the outcome.

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